CODY / VIBEWARE



CCP Communication Atlas

Local pairing, semantic events and the proposed private cloud connection.

DEVELOPMENT REFERENCE / 09 November 2026

Umberto Giacobbi / umbertogiacobbi.biz / Copyright (c) 2026

Current behavior and limitations. Original correction dates and screen IDs remain in the final register.

Contents

Use screen IDs for product identity; PDF page numbers are navigation only.

Current local connection	3
Local pairing and recovery	4
Messages and delivery	5
Cloud trust model	6
Cloud pairing ceremony	7
Cloud operation and limits	8
Voice and operator boundaries	9
Corrections register	10
Security	15

Current connection: Cody and its paired PC

Current repository source is the reference. Firmware execution, physical behavior and the private cloud service require their own verification. The device-first image presents a dashboard. Character artwork belongs to Studio/design work. Cloud firmware integration is deferred.

Cody

The canonical network/link owners authenticate the retained PC. The device model presents dashboard data and semantic events. Wi-Fi, PC presence and Codex app presence are separate facts.

CCP on the computer

One local broker owns the PC identity and active/pending credentials. Explicit tools create notices or questions; genuine hooks provide chat/turn activity. The agent does not receive device link keys.

Local transport

CCP1 uses UDP on the reachable private LAN. The link has authenticated message checks and current-presence challenges. Ordinary payloads are not encrypted. Do not publish these UDP ports to the Internet.

Independent cloud work

The new private relay is a separate service under implementation. It will require a distinct enrollment, device token and client grant. Existing local pairing does not automatically grant cloud access.

Sources: docs/cody/navigation.md, layout.md, runtime.md; plugins/cody-codex-plugin/README.md and protocol/CCP1.md.

Pair once, prove current presence again

Local workflow / original screens 12-15, 40-42. IP addresses and names are not credentials.

Start deliberately

BOOT > Options > Connections > Connect Codex. Keep link resumes the remembered PC. Pair again explicitly permits replacement. Forget opens a separate confirmation.

Read the intended display

Use all six characters from the device's current code after discovery. Code preparation is not measured progress. The pairing window is five minutes; no code is logged or broadcast in plaintext.

Persist both sides

PC and Cody stage separate candidates while retaining active records. Confirmed persistence and authenticated ACTIVE/COMMITTED complete association. A pending record alone is not success.

Recover real state

Ordinary LAN loss or an IP change resumes the saved association. Cancel preserves the old operational link. A local Forget does not revoke the independent PC record. Storage failure requires explicit repair.

Sources: docs/cody/navigation.md, layout.md, runtime.md; plugins/cody-codex-plugin/README.md and protocol/CCP1.md.

A notification is an event; an answer needs delivery

Semantic content carries stable identity, freshness and the actual outcome.

Activity

Follow the latest authorized prompt with exact chat/turn identity. Tools and completions in another chat cannot take over. A response-ready event does not prove task success or passed checks.

Questions

26 opens a question; 27 shows numbered rows, at most two visible. Read the full label/detail before Send. Selecting, scrolling or going Back does not answer.

Acknowledgements

PC receipt and recipient delivery are different. Only the recipient's authenticated acknowledgement permits 31 / Reply sent. The deadline includes delivery; timeout never chooses a reply.

Data discipline

Keep bounded printable ASCII, immutable request identities, original timestamps and explicit unavailable/error state. Native Codex approvals stay on the PC. A remote notification cannot authorize a privileged action.

Sources: docs/cody/navigation.md, layout.md, runtime.md; plugins/cody-codex-plugin/README.md and protocol/CCP1.md.

Cloud relay: three separate identities

Architecture under implementation. No live service or cloud-capable firmware is claimed.

Owner

Signs in through the configured identity provider. Claims the intended device and grants one client access.

Dot / client

Uses a scoped access token or a separately issued connection key. The server resolves its allowed devices.

Cody

Initiates HTTPS polling. A separate revocable token is bound to its P-256 key through DPoP proof.

Private relay / identity + explicit grants + bounded delivery

A device ID is an address, not permission. Every send, poll and acknowledgement checks the authenticated principal and its live grant.

One relay instance / state, cookie protection keys and bounded logs on an OS-mounted SMB/Samba share

ASP.NET Core + Aspire + OpenTelemetry; HTTPS polling first. Aspire/DCP has separate OS temporary-file and IPC requirements. Relay storage needs no Azure SDK or provider API. Hosting mode, Windows/Linux mount behavior and integration remain pending; no deployment is claimed.

Connect the person, the device and one client

Proposed cloud ceremony / a separate permission for each association.

1 / Device starts

After an explicit local action, Cody creates a pending enrollment bound to its device key and shows a short one-use code. No client grant exists yet.

2 / Owner claims

The owner signs into the relay's browser UI and enters the displayed code. The claim expires and is attempt-limited. Guessing an ID or finding an endpoint grants no access.

3 / Confirm on Cody

The physical confirmation must match that exact key-bound transaction and intended owner. The proposed firmware confirmation UI and protected credential storage remain to implement.

4 / Grant Dot access

The owner selects the already owned device and authorizes a particular client and scope. The server issues a separate connection credential or accepts the configured OAuth client grant. Other clients need their own grant.

The server proves control of enrolled credentials and a confirmed ceremony; it cannot continuously prove who physically holds the device.

A private service using a standard mounted share

Implementation target: ASP.NET Core .NET 10, Aspire and OpenTelemetry. Deployment is not part of this edition.

Relay persistence on SMB/Samba

One relay instance stores bounded JSON state, cookie protection keys and bounded operational logs on the OS-mounted share, with no local persistence fallback. The OS handles mount credentials. Local .env.local is configuration only. The relay uses no application database, Azure SDK or provider storage API.

HTTPS polling first

Cody initiates outbound HTTPS polling and acknowledges admitted events. The relay bounds queued content, expiry and duplicate identities. No inbound device port or WebSocket transport is required for the first version.

Windows and Linux mount gates

Verify exclusive locking, durable flush, atomic replacement and disconnect/crash handling on the actual mount for each host. No durable-write or failover guarantee follows from the design alone. A small VM on Azure or another host is an option; multiple writers are not supported.

Aspire and OpenTelemetry

Aspire/DCP has separate OS temporary-file and IPC requirements even with the dashboard disabled. OpenTelemetry keeps bounded relay logs on the share and supports optional remote OTLP, without a local telemetry database. Exclude secrets and message bodies. Aspire for development only or also for hosting remains undecided. Runtime verification is pending.

Sources: docs/cody/navigation.md, layout.md, runtime.md; plugins/cody-codex-plugin/README.md and protocol/CCP1.md.

Use the operation that the current source supports

Local customer controls, optional voice and engineering access have separate admission.

Voice is optional

The default device-first profile omits microphone acquisition. A voice-enabled build requires Dictation on and a current paired LAN link. Record voice or a dashboard double tap starts the documented route.

Audio evidence

Recording, encrypted transport, queue acceptance and recipient transcription are separate outcomes. The retained 15 October synthetic queue result did not prove speech receipt. No new physical voice evidence is claimed here.

Settings and diagnostics

Authenticated commands retain identity, nonce and actual acknowledgements. Diagnostic measurements come from canonical owners. Acknowledged settings do not prove physical pixels or sound.

Reset and approval

Restart and factory reset are separate, explicit actions. A reset ACK establishes admission; reboot and deletion need actual evidence. Codex's native approval remains on the computer.

Sources: docs/cody/navigation.md, layout.md, runtime.md; plugins/cody-codex-plugin/README.md and protocol/CCP1.md.

Corrections register

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
09 October 2026 **ERRATA**	**Saved state and reversible setup** Affected: Navigation originals 02, 04, 16, 25, 28-29; Planning originals 27-29; screens 04-16, 35B, 36, 40-42. The original analysis omitted a configured network without a Codex association and a reversible setup entry. Screens 40, 41 and 42 identified those distinct routes. Saved pairing, network recovery and replacement consent remain separate. The current device-first entry is BOOT > Options > Connections, replacing the character-hold instruction. Evidence / limits: Source/design evidence only; physical behavior was not established by this correction.
09 October 2026 **ERRATA**	**Standby recovery and malformed touch evidence** Affected: Navigation originals 28, 31-34; Planning originals 30-34; Service 14-18 / Connection 18-22; screens 04-06, 12, 35B, 41. A 90-second passive capture observed LAN/DHCP, Welcome inactivity return and standby, but no Connect Codex press. A separate capture recorded Yes at 16497 ms, Connect Codex at 16498 ms, then fifteen FF report bytes and ACK=FF at 17112 ms; expected ACK was AB. ESP_ERR_INVALID_RESPONSE stopped rendering. The correction separates an optional runtime acquisition error from rendering, interrupts gestures and requires a real release. Evidence / limits: These are retained measured failures. The FF hardware cause and corrected physical behavior were not established by those captures.

Corrections register

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
09 October 2026 **ERRATA**	**Action layout and touch admission** Affected: Navigation/Planning original 34; screens 05-15, 26-27, 35B, 36, 40-42; selected-multi-action.png and selected-single-action.png. The approved design uses two side-by-side rounded gradient cards, a centered third action, or one wide capsule. Earlier stacked-blue examples were superseded. Current captions are text-only; the older icon-left prescription is no longer the action contract. Question choices now use numbered single-line rows with full review before Send. Evidence / limits: Approved design references are not device captures.
10 October 2026 **ERRATA**	**Automatic chat context and freshness** Affected: Navigation originals 35-38; Planning originals 35-38; screens 15-20, 18U, 26-27, 31-32. CCP follows the latest authorized UserPromptSubmit with exact chat/turn identity. Manual Choose/Pick a chat is removed. Background events cannot select context; a tab click is not a documented focus event. Pairing, authenticated presence, app presence and real work remain distinct. Eight seconds without packets expires device state; active work becomes unavailable after 120 seconds without another hook. Evidence / limits: The owner confirmed Connected on 10 October for the preceding pairing implementation. That observation does not prove later hooks or presentation.

Corrections register

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
10 October 2026 **ERRATA**	**Retain, replace, cancel and forget** Affected: Navigation/Planning originals 39-42, 48; screens 12-16, 32, 35B, 40-42. Separate pending and active records on both peers. Persist before confirmation and promote only after authenticated ACTIVE/COMMITTED. Cancel preserves the old operational link; a pending PC candidate requires a new visible code and explicit replacement. Forget removes Cody's CCP records, not the independent PC record. Temporary disconnection does not authorize re-pairing. Evidence / limits: Source/design evidence only; physical behavior was not established by this correction.
10 October 2026 **ERRATA**	**Reply delivery and command outcomes** Affected: Navigation/Planning originals 43-46; Service 25-28 / Connection 29-32; screens 21-27, 31-32. Receipt is not recipient delivery. An immutable question ticket expires after 180 seconds including delivery acknowledgement. Timeout, offline, Back and Cancel never select an answer. Reply sent requires recipient acknowledgement. Restart retains settings; factory reset requires explicit confirmation and verified deletion. A command ACK proves admission, not reboot completion. Evidence / limits: Source/design evidence only; physical behavior was not established by this correction.

Corrections register

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
10 October 2026 **ERRATA**	**Boot artwork and current controls** Affected: Navigation/Planning originals 47-50; Manga original 02 and 28-31; screens 01-15, 36, 39-42. The complete 42-frame VibeWare artwork retains its authored 3.5-second dwell. The 10 October correction replaced the earlier title-free/twofold composition with a title, subtitle and 192 x 219 artwork at (137,164). It also removed the connection detour, clarified the five-minute pairing window and required successful settings write/readback. Current device-first routes are now integrated in the main body. Evidence / limits: Coordinates are source layout values. Schematics are neither simulator frames nor device captures.
11 October 2026 **ERRATA**	**English throughout** Affected: Navigation originals 01, 05-10, 29, 31, 34, 36, 38-46 and 53; Planning originals 01-02, 09-14, 16, 21-22, 39-46 and 53; Service 33 / Connection 37. Translate headings, diagrams, captions, prose and footers into English while preserving original correction dates, screen IDs, technical values and evidence limits. Evidence / limits: Document language and layout review only.
12 October 2026 **CHANGE**	**Navigation within the documents** Affected: All atlas pages; Manga prior final register. Coordinated document identity, navigable contents and bookmarks retain original screen references when pagination changes. The quickstart keeps its own four instructional A5 pages. Evidence / limits: Document authoring only.

Corrections register

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
09 November 2026 **ERRATA**	**Current device and private cloud boundaries** Affected: Current Navigation/Planning bodies; former Navigation pages 03-19 and Planning pages 03-27; map and A5 instructions. Replace character-led current routes with the device-first dashboard, BOOT Options, current carousel order, Timer, Quick guide and explicit reset screens. Preserve character artwork as Studio/design material. Consolidate corrections into this register and remove accumulated build/code history. The private cloud service is under implementation; cloud firmware and Dot integration are pending. Evidence / limits: Read-only source review and PDF authoring. No firmware build, test, simulator, device operation or cloud deployment accompanies this edition.
12 October 2026 **ADDENDUM**	**Operator and diagnostics boundaries** Affected: CCP operator/diagnostic routes; original diagnostic screen 38. Separate customer operations, authenticated diagnostics and USB engineering. A worker acknowledgement establishes admission or application of a command; the intended physical outcome requires separate observation. Debug and Release share the diagnostic entry without exposing a build-mode choice. Evidence / limits: Source review and document authoring only.
15 October 2026 **ERRATA**	**Audio queue admission did not prove speech receipt** Affected: Communication audio delivery / original voice screens 28-30. A synthetic localAudio item entered and left the public experimental queue, while Desktop reported that audio content was omitted because audio input was unsupported. Queue acceptance is not transcription or successful recipient audio delivery. Evidence / limits: Observed synthetic queue/Desktop result. Physical-device voice submission was not performed.

Security

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
10 October 2026 SECURITY	**Fresh proof before presence or current data** Affected: CCP FIND / READY / COMMITTED; original screens 12-15, 18U, 26-27, 31-32. FIND alone cannot establish presence, redirect notifications or release current data. READY binds a fresh challenge to the observed device endpoint; COMMITTED proves the matching response. Distinct active/pending records, strict field bounds, nonces and one-use challenges preserve explicit admission. Firmware/plugin changes must agree; no legacy handshake fallback is claimed. Evidence / limits: Source-based correction; neither runtime adversarial behavior nor physical recovery is certified.
10 October 2026 SECURITY	**Bound work and keep residual risks explicit** Affected: Firmware UDP owner; PC discovery/pairing; ACTIVITY / TECH / EVENT; optional HTTP. Global/per-source ingress budgets and bounded receive slices precede expensive work. Discovery and pairing have finite packet/offer/work budgets. Broker network diagnostics are bounded; startup-only runtime.log retention remains separate. Six-character CCP transcripts permit offline guessing, and ordinary HMAC payloads are readable. Standard PAKE/high-entropy QR and authenticated encryption remain deferred. Optional plaintext device HTTP is a separate review scope. Evidence / limits: HMAC authenticates but does not provide confidentiality. No compromised-PC protection or production-security certification is claimed.

Security

Retained corrections and evidence; original dates, page references and screen IDs remain explicit.

DATE / TYPE	REVISION DETAILS AND SUPPORTING MATERIAL
07 November 2026 **SECURITY**	**Account projections stay bounded and authenticated** Affected: CCP ALLOWANCECAP / ALLOWANCE / ACCOUNT and later bounded account projection; Planning original 24; Communication core 03 and 09. The current endpoint, nonce and authenticated association bind account projections. Credentials, account identity email and message contents stay on the host. Source age, unavailable samples and separate account generations prevent stale or mixed-account graphs being shown as current. Evidence / limits: Source ownership only; this document does not establish live account login or end-to-end device delivery.
09 November 2026 **SECURITY**	**Cloud authorization is a separate trust boundary** Affected: New cloud relay concept; original local pairing screens 12-15 remain local. No new firmware screen IDs are assigned. Use owner sign-in, a key-bound device credential and an explicit owner/device/client grant. An ID, LAN pairing key or account label never authorizes arbitrary destinations. A short enrollment code is one-use and expires; owner claim and physical confirmation belong to the same transaction. HTTPS protects transport; DPoP binds the device token to its P-256 key. Revocation is checked server-side. Existing LAN UDP is not exposed publicly. Evidence / limits: Private implementation is in progress. Build, runtime/security review, Dot integration, firmware storage and physical confirmation are pending.